

Barath Suresh

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EDUCATION

Arizona State University – Tempe, AZ

Master of Science in Computer Science | GPA: 3.67/4.00

August 2025 – May 2027

SASTRA Deemed University – India

Bachelor of Technology in Computer Science

August 2020 – May 2024

SKILLS

Languages: Java, Python, Go, TypeScript, SQL, C/C++

Backend & APIs: Spring Boot, Spring Security, Spring AI, FastAPI, REST APIs, Microservices, Distributed Systems, JWT

AI / ML: Retrieval-Augmented Generation (RAG), pgvector, PyTorch, LoRA/QLoRA, NVIDIA NIM, Prompt Engineering

Cloud & DevOps: AWS (Lambda, Kinesis, SageMaker), GCP (Cloud Run), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions

Data & Observability: PostgreSQL, Redis, Hazelcast, RabbitMQ, Kafka, Prometheus, Grafana, Zipkin, Distributed Tracing

WORK EXPERIENCE

TATA ELXSI LTD

December 2024 – July 2025

Software Engineer

Chennai, India

- Engineered an asynchronous bulk-import pipeline using Java, Spring Boot, RabbitMQ, and PostgreSQL for OTT content catalog metadata, eliminating UI timeouts and reducing processing time by 70% on 20,000-row files
- Secured 50+ REST APIs with Spring Security, Azure AD OAuth 2.0, and JWT, cutting auth latency from 100ms to 7ms
- Built a log anomaly detection pipeline across 6 microservices using Python, Spring AI, and Gemma 3 to flag batch-upload crashes, cutting incident detection time from 2 hours to 40 minutes
- Streamed microservice crash logs through RabbitMQ into Grafana Loki for real-time triage, and added a Hazelcast cache (Spring Cache) for hot catalog reads

TATA ELXSI LTD

January 2024 – June 2024

Software Engineer Intern

Bengaluru, India

- Built high-throughput telemetry ingestion pipelines with Python, FastAPI, and MQTT for streaming set-top box metrics
- Built WebSocket relay services and PostgreSQL persistence layers for real-time Grafana dashboards

PROJECT EXPERIENCE

THROTTLR: Distributed Rate Limiting Platform | [GitHub](#) | Java, Spring Boot, React, Redis, Lua, GCP Cloud Run, Terraform

- Engineered a distributed rate-limiter on GCP Cloud Run using atomic Redis Lua scripts across multiple algorithms to protect APIs and LLM inference endpoints from traffic spikes
- Deployed via Terraform, tuning autoscaling and CPU concurrency to boost throughput 11.2x (9.8 to 109.5 req/sec) across 20K requests with zero errors and 182ms mean latency
- Built a React and TypeScript dashboard for live throttle-rule configuration and client usage analytics

PRISM: Distributed Video Streaming Platform | [GitHub](#) | Java, Spring Boot, WebFlux, RabbitMQ, Docker, AWS S3, Zipkin

- Architected 8 Spring Boot microservices where RabbitMQ decouples uploads (up to 5GB) from FFmpeg transcoding into 6 multi-bitrate HLS tiers, cutting upload latency 70%
- Built a reactive Spring WebFlux proxy for 5x more concurrent streams, with multi-tenant isolation via BCrypt API keys and JWT scopes enforced in Spring Cloud Gateway
- Traced all services with Zipkin, Prometheus, Grafana, and Loki, cutting debugging time 60%

SIFT: Semantic Codebase Search (Code RAG) | [GitHub](#) | Python, pgvector, tree-sitter, NVIDIA NIM, Pytest

- Architected a code RAG engine using tree-sitter AST parsing, NVIDIA NIM embeddings, and pgvector to answer repository-wide questions with file and line citations
- Implemented hybrid search with Reciprocal Rank Fusion (RRF) for exact symbol and natural-language retrieval
- Added a bounded thread pool with incremental caching, boosting embedding throughput 3x on a 150K-chunk benchmark and cutting full re-index time from ~36 minutes to ~6 seconds

NEUROSCRIBE: MRI-to-Report AI Pipeline | [GitHub](#) | PyTorch, MONAI, NVIDIA NIM, QLoRA, SageMaker, Terraform

- Fine-tuned LLMs with QLoRA and PEFT to generate radiology drafts from MRI scans, lifting ROUGE-L from 0.49 to 0.75 and BERTScore from 0.88 to 0.96 over the base model
- Added segmentation-grounded filtering, reaching 98.3% factual groundedness on 119 held-out examples with zero hallucinated measurements
- Deployed to AWS SageMaker endpoints with Terraform